ALEXIS at AlexandriaX-2026 Subtask 2: Capacity Discovery and Calibrated Preference Optimization for Cross-Dialect Arabic MT

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Abstract

Cross-dialect Arabic machine translation asks for translation between Modern Standard Arabic and regional dialects with almost no parallel text. We present a system for Subtask 2 of AlexandriaX-2026 (El Mekki et al., 2026), built on two findings about where the gains come from. First, data was not the binding constraint: once our mined-and-synthetic recipe had decelerated to +1.8% relative spBLEU per round, jointly raising LoRA (Hu et al., 2021) rank and epoch budget beat three further data-tuning rounds combined (+8.9%); we did not vary the two independently, so we cannot attribute the gain to rank alone. Second, planned epoch budgets were consistently too short, and training past them was consistently the largest lever we found. We fine-tune NLLB-200-distilled-600M (NLLB Team et al., 2022) on a three-tier corpus of real, mined and synthetic pairs, then refine it with Direct Preference Optimization (Rafailov et al., 2023). Ablating three reward designs isolates a third finding: rewarding agreement with self-distilled references regresses real quality despite healthy training dynamics. Restricting the reward to references we trust, with a reference-free substitute for the two zero-resource dialects, yields a real gain that a further iteration gives back. Every configuration was scored on the official leaderboard. Our submission reached spBLEU 25.84 and chrF++ 39.02, ranking **2nd of 4** teams; post-evaluation experimentation reached 28.04.

1 Introduction

Arabic dialects are overwhelmingly spoken or informally written, so almost none of the 20-plus source-target directions Subtask 2 requires have an existing parallel corpus. The task translates short banking-support sentences between Modern Standard Arabic (MSA) and six dialects – Egyptian, Lebanese, Moroccan, Palestinian, Saudi and Tunisian – scored with spBLEU and chrF++ on a blind test set of 3,574 rows.

The obstacle is not translation difficulty in the usual sense; the sentences are short and formulaic. It is data scarcity, and it is distributed unevenly. Two dialects, Egyptian and Lebanese, have no real or minable parallel text at all, and together they account for roughly 39% of the test rows.

Findings. Our system is organized around three findings, each identified by trusting real Codabench feedback over offline proxies whenever the two disagreed:

- **Capacity looked more binding than data.** Three rounds of widening a mined and synthetic data recipe produced monotonically shrinking gains (+3.8%, +2.7%, +1.8% relative spBLEU). Jointly raising LoRA rank and epoch budget then outperformed all three combined (+8.9%); since we changed both hyperparameters at once, we read this as evidence that the data recipe alone had plateaued rather than as a clean isolation of capacity from training length, and we redirected the project accordingly.
- **Planned training budgets were too short.** Whenever validation loss had not actually plateaued, continuing training past its originally scheduled epoch count was the largest single lever available – including after we switched from cross-entropy to preference optimization.
- **DPO helped only under a specific, non-obvious reward design.** A first attempt that trusted every data tier degraded real quality; diagnosing why led to a second attempt that produced our best result; a third, which naively repeated that recipe on its own output, regressed again.

Section 2 describes the task and data. Section 3 walks through the system's four development

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phases and the preference-optimization stage. Section 4 reports the experimental configuration and Section 5 the official and post-evaluation results, including the DPO ablation. Section 6 concludes, and Section 7 discusses limitations. Our code is publicly available[1].

2 Background

2.1 Task

Each test instance is a `(source dialect, source sentence, target dialect)` triple. MSA appears only as a source or pivot, never as a scored target. Systems output one translation per row, scored with SentencePiece-tokenized BLEU (spBLEU, via the FLORES-200 tokenizer (NLLB Team et al., 2022)) and chrF++ (Popović, 2017) against held-out references, following standard practice for low-resource and dialectal MT evaluation (Post, 2018). The 3,574-row blind test set covers all six target dialects (Egyptian, Lebanese, Moroccan, Palestinian, Saudi, Tunisian). We do not have the exact per-dialect row counts for the released test set; the 39% figure we report for Egyptian and Lebanese combined (Section 1) is our own estimate from inspecting the test file, not an official breakdown.

2.2 Data

No public parallel corpus covers this dialect set directly, so we built a three-tier training corpus from ArBanking77 (Jarrar et al., 2023),[2] a public multi-dialect extension of the Banking77 intent-classification dataset. The *gold* tier holds 16,000 real, human-written MSA↔Palestinian pairs and is the only tier with a genuine ground-truth target. The *silver* tier holds 28,668 cross-dialect pairs mined by embedding similarity within a shared banking-intent bucket – same topic, not a verified translation – covering MSA, Palestinian, Moroccan, Saudi and Tunisian. The *bronze* tier holds 25,471 synthetic pairs for Egyptian and Lebanese, generated by backtranslating through the model itself and filtered for degenerate output. The final corpus totals 70,139 examples across 38 direction pairs (Table 3, Appendix B). Tier caps were tuned iteratively (Section 3) rather than fixed up front.

[1] https://github.com/Momen-Ibrahim-Fawzy/ALEXIS-AlexandriaX-Subtask2-CrossDialectMT

[2] https://github.com/SinaLab/ArBanking77

2.3 Related Work

We build on NLLB-200 (NLLB Team et al., 2022). FLORES-200 provides a dedicated code for MSA and for several individual Arabic varieties, but not a one-to-one code for every dialect in this task: our Palestinian and Saudi directions use the closest available broader Levantine and Peninsular Arabic codes rather than dialect-specific ones. We adapt NLLB-200 first with LoRA (Hu et al., 2021) and later with full-parameter fine-tuning. For preference optimization we follow DPO (Rafailov et al., 2023); prior work applying it directly to MT quality (Xu et al., 2024) motivated treating it as a lever distinct from further supervised fine-tuning, which Section 5 confirms. Our negative model-averaging result (Section 3) tests model soups (Wortsman et al., 2022) on checkpoints of unequal quality. Work on Arabic financial and banking-domain NLP, including the AraFinNLP shared task (Malaysha et al., 2024), provides the broader domain context for this data; two systems built directly on ArBanking77 for cross-dialectal intent detection (Ramadan et al., 2024; Elkordi et al., 2024) face the same dialect-coverage gap we do. Quantifying dialectal distance with contrastive learning (Sakr et al., 2024) and extending a corpus via reverse translation to offset missing dialect data (Tahssin et al., 2020) are both directly relevant to how we handle Egyptian and Lebanese.

3 System Overview

Every experiment adapts the same backbone, `facebook/nllb-200-distilled-600M`. The system was built in four phases, each begun because the previous one had stopped paying off; Section 3.6 covers the preference-optimization stage layered on top.

One prerequisite governs all of them. NLLB expects inputs of the form `[lang_id, content_ids..., eos_id]`, and the fast-tokenizer wrapper does not reliably produce them: called with `src_lang=...` it silently omits the FLORES-200 source-language token, and appends a spurious trailing token otherwise. We build the sequence manually and identically at training and inference time. Every number below assumes this fix.

3.1 Phase 1: Data-Recipe Tuning (LoRA)

From a LoRA baseline (rank 16, 3 epochs), three rounds widened the silver and bronze mining caps,

grew the gold tier from 2,500 to 16,000 pairs, and corrected a tokenization artifact in the Tunisian rows (Appendix A). Every round helped, and every round helped less than the one before it: +3.8%, +2.7%, then +1.8% relative spBLEU. Extrapolating that curve, another round of data tuning was not worth its cost, so we changed variable instead.

3.2 Phase 2: Capacity Discovery

Holding the data recipe fixed, we raised LoRA rank (32→64) and the epoch budget (5→8) together, changing nothing else in the data pipeline. That single step gained +8.9% relative spBLEU – more than the three data rounds combined. We did not ablate rank and epoch budget separately here, so we cannot say how much of this gain is attributable to capacity as opposed to training length alone; what the result does show is that the data recipe was no longer the binding constraint, which is why the rest of the project moved away from further data tuning.

3.3 Phase 3: Full Fine-Tuning

Carrying the capacity result to its conclusion, we merged the best LoRA adapter into the base weights and fine-tuned all parameters. The first attempt trained nothing at all: `peft` leaves the merged weights frozen, and the run completed without error or warning, producing a checkpoint byte-identical to its initialization (Appendix A gives the mechanism and the fix, which generalizes to any `peft`-merge-then-full-finetune pipeline).

Once training genuinely ran, an unplanned comparison became available. Two otherwise identical runs differed only in which adapter initialized them: one trained on original bronze data, one on self-distilled bronze. The self-distilled run scored *better* offline and worse on the leaderboard. Neither dev set covers Egyptian or Lebanese, so the offline metric could not see what the self-distillation had cost – a blind spot that reappears in Section 3.6.

3.4 Phase 4: Training Duration as a Lever

Every full fine-tuning run's validation loss was still falling when its epoch budget expired, which says more about scheduling budgets in advance than about the data. Continuing the Phase 3 run two epochs past its planned three gained +0.71 spBLEU over those two additional epochs combined (Table 4), the largest single-checkpoint gain to that point; our logging does not isolate the contribution of each epoch individually. We then dropped fixed budgets entirely: three further rounds of up to 10 epochs each, restarting from the previous round's best checkpoint with a fresh optimizer state, early-stopped on two-epoch validation-loss patience. Cumulatively this was the largest gain of the project, with returns diminishing and turning slightly negative by the third round (Section 5).

3.5 Model Soups

Separately, we tested greedy model soups (Wortsman et al., 2022) across independently initialized lineages. Every candidate weaker than the current best degraded the average once mixed in, so the greedy procedure retained the single best checkpoint alone and the soup reduced to its own input. We report this as a validated negative result rather than a bug: across checkpoints of unequal quality, averaging bought nothing.

3.6 Preference Optimization

Shared protocol. Once continued fine-tuning plateaued, we applied DPO to a Phase 4 checkpoint (eval loss 0.7228; neighboring checkpoints in the same run scored 27.37–27.59 real spBLEU). All three configurations start from it and share one sampling protocol: draw $K{=}4$ candidates, add the training reference to the pool, score every member, and pair the highest- and lowest-scoring members as chosen and rejected. They differ only in how that score is computed.

3.6.1 Configuration A: Uniform Reward

Sentence-level chrF++ against the training reference plus a dialect-marker-authenticity bonus, applied identically across all six dialects ($\beta{=}0.1$). Preference accuracy climbed smoothly to 0.68; real spBLEU fell to 25.02, against a then-best 27.89. Two flaws compounded. The reward treated self-distilled bronze text as ground truth for Egyptian and Lebanese – the Phase 3 trap, now written into the objective. And with a real reference in the pool 99% of the time, the top-scoring candidate was usually that reference, so DPO largely re-taught what cross-entropy had already taught rather than adding to it.

3.6.2 Configuration B: Trust-Restricted Reward

From the same base checkpoint, B addresses each flaw directly. Bronze text is excluded from the reward for the four dialects that have gold or silver data. Egyptian and Lebanese, which have nei-

ther, receive a reference-free reward instead: BGE-M3 (Chen et al., 2024) cosine similarity to the source, plus the marker bonus. And β rises to 0.3, holding the policy nearer its reference model. The reward now scores only signal the model cannot obtain by memorizing its training targets. B gave our best result overall, spBLEU 28.04, above every full fine-tuning checkpoint in the same lineage.

3.6.3 Configuration C: Second Iteration

C repeats B's recipe on B's own output: same reward, initialized from B, with fresh candidates sampled from the improved policy. It regressed on every offline metric tracked and was not submitted (Section 5). We want to flag this decision explicitly: we withheld C from submission on the strength of offline evidence alone, in the same paper that argues offline signal can be misleading (Section 7). We judged the offline regression here to be large and consistent across every tracked direction, unlike the borderline cases discussed there, but we did not verify C on the real leaderboard, so we cannot rule out that it would have performed better than our offline estimate suggests.

4 Experimental Setup

LoRA runs used rank $\in \{16, 32, 64\}$, up to 8 epochs and learning rate 1e-4. Full fine-tuning used per-device batch size 2 with gradient accumulation 16 (effective batch 32), learning rate 3e-5, bf16, gradient checkpointing, and early stopping with patience 2 once budgets moved from fixed to open-ended. DPO used learning rate 1e-5, $\beta \in \{0.1, 0.3\}$, batch size 6, 7 epochs, and fp32 precision throughout; Appendix A details a silent-failure mode this last choice avoids.

Decoding used beam search (beam 4), `no_repeat_ngram_size=3` and repetition penalty 1.3, guarding against the degenerate repetition loops seen in roughly 0.2% of early predictions.

All runs used two dev sets for offline model selection: *gold_dev* (real, leak-free MSA↔Palestinian pairs) and *silver_dev* (20 mined directions). We treat both as directionally but not proportionally reliable relative to the real leaderboard (Section 7). Training ran on single-GPU NVIDIA instances; at 600M parameters, both full fine-tuning and DPO fit comfortably in fp32 without optimizer-state offloading.

Result	spBLEU	chrF++
Official (rank 2/4)	25.840	39.019
Best post-eval. (DPO Config. B)	**28.042**	**40.774**

Table 1: Official vs. post-evaluation results (Section 5).

5 Results

Our official submission scored spBLEU 25.840 and chrF++ 39.019, ranking 2nd of 4 teams: 2.56 spBLEU behind 1st place (28.401) and 1.31 ahead of 3rd. Everything reported beyond this point is post-evaluation – run and analyzed after the deadline, not eligible for the official ranking – and is marked as such throughout. Our best post-evaluation checkpoint closes the gap to 1st place to 0.36 spBLEU. Table 1 and Figure 1 summarize the full trajectory of real (never offline-estimated) spBLEU across the project.

5.1 DPO Ablation

Table 2 and Figure 2 report all three reward configurations. The A→B reward-design fix is the whole effect: the two share a base checkpoint, optimizer, epoch budget and architecture, and differ only in which data tier is trusted as a reward reference and how strongly (β) the policy is held near its reference model.

Training-time diagnostics did not predict any of this. Configurations A and C both show smooth, monotonically climbing preference accuracy, indistinguishable in shape from B's, yet A regressed sharply and C regressed on every offline metric tracked. The reason is straightforward once stated: preference accuracy measures only how well a policy satisfies its own reward, and says nothing about whether that reward corresponds to real translation quality.

We take this as the paper's central methodological finding. For MT preference optimization over data tiers of unequal trust, reward auditing is not optional, and a successful configuration does not automatically compound under further iteration.

5.2 Do the Two Objectives Compound?

We also continued plain full fine-tuning (cross-entropy) from the Configuration B checkpoint for further epochs, to test whether the two objectives stack. Offline metrics were mixed (two of three dev signals improved) and the real result came in at 27.51, below Configuration B's 28.04. We read this as plain fine-tuning partially undoing DPO's

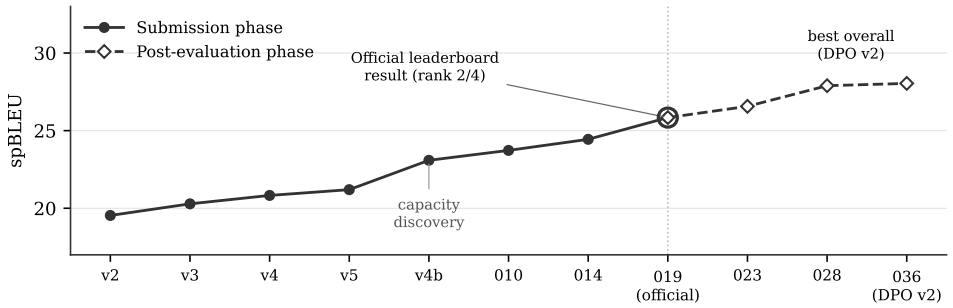


Figure 1: Real Codabench spBLEU across the project. Filled circles: submission phase; open diamonds: post-evaluation.

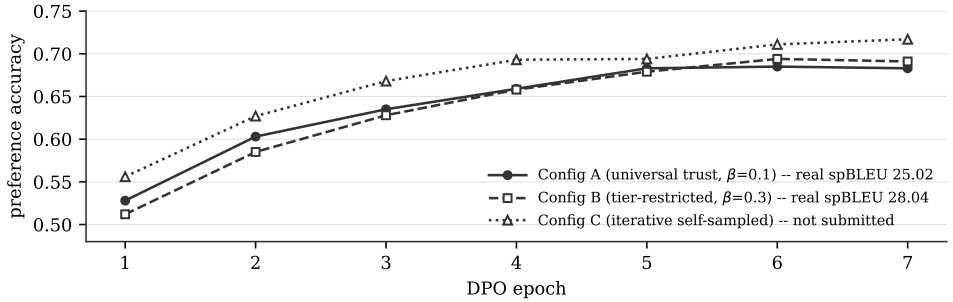


Figure 2: Preference accuracy by DPO epoch, all three configurations. Training dynamics alone do not distinguish the configuration that helped (B) from the two that hurt (A, C).

Config.	Pref. acc.	Real spBLEU	Post-eval.?
A	0.683	25.02	Yes
B	0.691	**28.04**	Yes
C	0.717	offline regressed	No

Table 2: DPO configurations: final-epoch preference accuracy vs. real outcome. None of A, B, or C is the official ranked submission – that is the Phase 4 checkpoint in Table 1 (spBLEU 25.840). "Post-eval.?" marks whether a configuration was scored on Codabench after the official deadline, in the post-evaluation phase. A and B share an identical base checkpoint (neighboring same-lineage checkpoints scored 27.37–27.59); C continues from B itself. C's offline silver_dev score dropped 5.6 points relative to B across every direction tested; given the reliability of *directional* offline signal established in Section 7, we withheld it from Codabench entirely rather than scoring it post-evaluation.

targeted fix rather than building on it, which is what one would expect from continuing to optimize the very cross-entropy objective the reward was designed to move beyond.

6 Conclusion

Team ALEXIS placed 2nd of 4 on AlexandriaX-2026 Subtask 2 with spBLEU 25.84, reaching 28.04 in post-evaluation experimentation. Two lessons generalize beyond this shared task. When a data recipe's returns decelerate, model capacity is worth testing as an independent lever before assuming more data tuning is the answer. And preference optimization for MT is only as good as the trust boundary drawn around its reward's reference data – a boundary that training dynamics alone cannot reveal, and that does not automatically compound under further iteration.

Future work includes replacing our lexicon and similarity proxy with a reference-free, learned quality-estimation reward, and testing whether the capacity lever from Phase 2 still holds at a larger backbone scale.

7 Limitations

Offline dev metrics were directionally but not proportionally reliable. Rankings between clearly different candidates transferred to the real leaderboard, but several close full fine-tuning rounds show real spBLEU deltas that offline eval loss under- or overstated, sometimes by more than the gap between competing submissions.

We have no variance estimate across random seeds; the submission budget was better spent on

distinct configurations than on repeats.

Our reference-free DPO reward for Egyptian and Lebanese is a proxy for adequacy, not a learned quality-estimation signal. We attempted to substitute a reference-free QE model (CometKiwi) but could not complete the download under this project's network constraints. This remains our most promising near-term extension.

Cold-start dialect quality still depends on synthetic bronze data drawn from the model's own earlier checkpoints. Our fixes reduce the self-distillation risk identified in Phase 3 but do not eliminate it.

Model and configuration selection drew on feedback from 24 real leaderboard submissions over the course of the project. This is more informative than a single held-out validation set, but it also means our final configuration was chosen with more exposure to the test-set signal than a strictly single-shot protocol would allow; some of our reported gains may not fully replicate under a cleaner selection procedure.

Finally, we report only aggregate spBLEU and chrF++ across all six dialects for each configuration; Codabench does not expose a per-dialect or per-direction breakdown of the official score, and we did not reconstruct one offline. This means we cannot directly verify from official results alone whether Configuration B's gain is concentrated in the two zero-resource dialects it specifically targets Egyptian and Lebanese or is spread more evenly across all six; our offline dev sets only partially cover this, since neither *gold_dev* nor *silver_dev* includes Egyptian or Lebanese directions.

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A Implementation Details and Reproducibility

This appendix documents three implementation-level issues encountered during development. Each carries a lesson that generalizes beyond this project.

Tunisian tokenization artifact (Phase 1). An audit of the raw Tunisian rows found that 29.4% carried a stray leading underscore introduced during data collection. Left uncorrected, the model reproduced the artifact in 35% of its Tunisian predictions. Stripping it before corpus construction was one of four data-recipe fixes applied in Phase 1 (Section 3).

Silent freeze on `peft`-merge-then-full-finetune (Phase 3). `peft`'s `PeftModel.from_pretrained` freezes every base-model parameter, and `merge_and_unload()` does not restore `requires_grad` on the merged weights. Our first full fine-tuning run therefore trained nothing for two full epochs: the resulting checkpoint was byte-identical to its initialization, confirmed by checksum, with no error or warning from the training loop. We fixed this with an explicit `requires_grad_(True)` call after merging, plus a trainable-parameter-count assertion at training start. We recommend that assertion as standard practice for any `peft`-merge-then-full-finetune pipeline.

bf16 precision floor in low-learning-rate DPO. bf16 DPO log-probability updates at learning rate 1e-5 can fall below the format's precision floor, silently making training a no-op with no visible error. We therefore trained all DPO runs in fp32 rather than bf16, and confirmed that preference accuracy moved as expected under fp32 (Section 4).

B Hyperparameters and Data Tiers

Tier	Examples	Trust
Gold (MSA↔Pal.)	16,000	Verified
Silver (mined)	28,668	Topic-matched
Bronze (synthetic)	25,471	Backtranslated
Total	70,139	38 directions

Table 3: Final training corpus composition.

Full fine-tuning otherwise used standard `Seq2SeqTrainer` defaults: linear warmup, weight decay 0.01, and best-checkpoint selection by eval loss. DPO used a linear warmup over 6% of steps and gradient clipping at norm 1.0. Full hyperparameters are in the released code.

C Techniques Tried and Rejected

Six early-phase techniques were tested and cleanly rejected by real leaderboard evidence. MSA-pivot decoding lost to direct decoding in 11 of 12 non-MSA directions. Diverse-beam MBR reranking improved 8 of 20 directions, at noise level. Cross-model consensus reranking failed because the weaker model was wrong more often, so trusting disagreement just meant trusting it more. A decode hyperparameter sweep found a best alternative beating the default by roughly 0.02 spBLEU. Checkpoint averaging underperformed the single best checkpoint. And exact-match retrieval, though 39% of test sources matched a training source verbatim, drew none of those matches from the gold tier, so substituting them risked doing harm.

D Full Submission History

Table 4 lists key milestones across the project's real, officially scored submissions; the complete

chronological log of 24 submissions is released with the code.

System Iteration	spBLEU	chrF++
Baseline LoRA (r=16)	19.536	35.010
Expanded LoRA (r=64)	23.088	37.440
Full Fine-Tune (init.)	23.727	37.730
Full Fine-Tune (+2 epochs)	24.440	38.259
FT, Extended (**Official**)	25.840	39.019
FT, Extended (ep. 12)	26.565	39.610
FT, Extended II (ep. 22)	27.894	40.538
DPO Config. A	25.017	39.136
DPO Config. B (**Best**)	**28.042**	**40.774**

Table 4: Key milestones from the full submission history, renamed here for readability; exact run tags and the complete chronological log (24 submissions) are released with the code. Rows below the bold official row are post-evaluation-phase.